Base Fine-tuned / Refined A: formula-cluster bootstrap 95% CI

A Magnetic-like candidates

MatterGen Base
MatterGen FT
Con-CDVAE Base
Con-CDVAE FT
Crys-JEPA Base
Crys-JEPA Refined
CrystalFormer Base
CrystalFormer FT
CrystaLLM Base
CrystaLLM FT

0.0 0.2 0.4 0.6 0.8 1.0

Fraction closer to magnetic than NM reference

B Magnetic-reference contrast

MatterGen Base
MatterGen FT
Con-CDVAE Base
Con-CDVAE FT
Crys-JEPA Base
Crys-JEPA Refined
CrystalFormer Base
CrystalFormer FT
CrystaLLM Base
CrystaLLM FT

−4 −2 0 2 4 6

Contrast: d(NM) − d(magnetic)

C Magnetic classes

AFM FM FiM

MatterGen Base
MatterGen FT
Con-CDVAE Base
Con-CDVAE FT
Crys-JEPA Base
Crys-JEPA Refined
CrystalFormer Base
CrystalFormer FT
CrystaLLM Base
CrystaLLM FT

0.0 0.2 0.4 0.6 0.8 1.0

Fraction among magnetic-like candidates

D DFT confirmation: Base groups

MatterGen Base
Con-CDVAE Base
CrystalFormer Base
CrystaLLM Base

0.0 0.2 0.4 0.6 0.8 1.0

Fraction confirmed magnetic by DFT (Wilson 95% CI)